PERSONAL INFORMATION

# Hai Jiang

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Gender Male | Date of birth 21 December 1997 | Nationality Chinese

JOB APPLIED FOR Ph.D. Applicant

## WORK EXPERIENCE

April 2026–Present

### Image Algorithm Engineer

Shenzhen Image Technology Co., Ltd.

- Develop C++ geometry algorithms for PCB copper-layer Gerber parsing and polygon feature extraction, converting raw contour data into structured inspection-ready representations.
- Extract edges, traces, corners, blocks, pads, and other manufacturing-critical geometric entities from large-scale polygon sets for downstream process inspection and quality analysis.
- Design computational geometry and topology analysis modules, optimize memory usage and throughput, and maintain unit tests and algorithm documentation for production use.

Skills C++, computational geometry, topology analysis, Gerber parsing, unit testing, technical documentation.

August 2025–April 2026

### Computer Vision Algorithm Engineer

Shenzhen Zhongrui Power Technology Co., Ltd.

- Built an automated AI engine for a property-management data platform using YOLO, enabling object recognition, segmentation, and classification from user-uploaded images.
- Integrated OCR for receipts, signs, and document-like scenes, transforming detected text into structured outputs for platform workflows.
- Packaged the inference pipeline behind standardized APIs for web integration, using ONNX export and TensorRT acceleration to improve real-time response latency.

Skills YOLO, OCR, ONNX, TensorRT, API integration, computer vision deployment.

July 2022–July 2025

### Research Assistant

Computational Medical Imaging Laboratory, Sun Yat-sen University

- Designed and validated algorithms for national medical AI projects covering MRI, dual-energy CT, and digital breast tomosynthesis; contributed to grant proposals, final reports, and industry collaboration.
- Developed an anatomy-guided multitask learning framework for MRI-based placenta accreta spectrum classification, achieving AUC $>$ 0.99 and first-author acceptance at ISBI 2025.
- Built and evaluated deep learning models for DECT lymph-node metastasis prediction and DBT multi-view detection, supporting peer-reviewed manuscripts and translational research outcomes.

Skills Medical imaging, deep learning, PyTorch, TensorFlow, data preprocessing, research writing.

## EDUCATION

**September 2019–July 2022** — **Master of Science in Computational Mathematics** — M. Sc.

Nankai University (NKU, Project 985 & 211, Double First-Class), Tianjin, China

Thesis: GANs-Based Personal Style Imitation of Chinese Handwritten Characters.

Developed an end-to-end CycleGAN framework achieving 85% visual similarity (10% improvement over baselines).

Key Skills: GANs, style-transfer learning, Python, PyTorch, data pre-processing.

Supervisors: Prof. Yunhua Xue, Prof. Chunlin Wu.

Relevant Coursework: Approximation Theory, Numerical Optimization, Functional Analysis, Matrix Computation, Numerical PDEs.

GPA: 74.9/100.0.

**September 2014–July 2018** — **Bachelor of Engineering in Information Security** — B. Eng.

Lanzhou University (LZU, Project 985 & 211, Double First-Class), Lanzhou, Gansu, China

Thesis: Improved Upper Bounds of Roman Domination Number in Maximal Outerplanar Graphs.

Focused on graph theory and combinatorial optimization.

Supervisor: Prof. Zepeng Li.

Relevant Coursework: Discrete Mathematics, Data Structures, Operating Systems, C/C++ Programming, Database Theory.

GPA: 4.15/5.00.

## RESEARCH EXPERIENCE

**July 2022–July 2025** — **Computational Medical Imaging Laboratory** — R. A.

School of Computer Science and Engineering, Sun Yat-sen University

Project: Placenta Accreta Spectrum Disorder Classification.

- Designed and validated algorithms for national medical AI projects covering MRI, dual-energy CT, and digital breast tomosynthesis; contributed to grant proposals, final reports, and industry collaboration.
- Developed a prior-knowledge-based deep learning approach for classifying PASD (binary classification task), achieving AUC of 0.9902 [6].
- Developed a multi-task learning model using T2-WI MRI images (multi-class classification task), achieving AUC of 0.80.
- Published: "Anatomy-Guided Multitask Learning for MRI-Based Classification of Placenta Accreta Spectrum and Its Subtypes" [1].

Skills: Literature review, data preprocessing, PyTorch, research writing.

**December 2023–January 2024** — **Computational Medical Imaging Laboratory** — R. A.

School of Computer Science and Engineering, Sun Yat-sen University

Project: Breast Cancer Metastasis Prediction.

- Designed a CNN-based framework using dual-energy CT scans to predict metastasis.
- Achieved AUC of 0.85 (mean cross-validation AUC).
- Manuscript submitted to MICCAI 2024 and under revision for Journal of Medical Physics [4].

Skills: TensorFlow, Keras, data analysis, experimental design, research writing.

**June 2023–November 2023** — **Computational Medical Imaging Laboratory** — R. A.

School of Computer Science and Engineering, Sun Yat-sen University

Project: Multi-view Learning for Atypical Architectural Disorder Detection in Breast DBT. [2]

- Contributed to the multi-view learning framework for atypical architectural disorder detection in digital breast tomosynthesis (DBT) images.
- Introduced a novel Siamese module with triplet loss to enhance feature extraction and detection performance.

Skills Mask-RCNN, Detection, Siamese Network, Multi-view, Triplet-Loss.

January 2022–June 2022 **Image Analysis Team**

School of Mathematical Sciences, Nankai University

Project ADMM Model for Compressive-Sensing MRI.

– Reproduced iterative mathematical equations from Deep ADMM-Net for Compressive-Sensing MRI using C++, Python, and PyTorch.

Skills Compressive-sensing theory, neural networks, MRI reconstruction.

## PUBLICATIONS

[1] **Hai Jiang**, Qiongting Liu, Yuanpin Zhou, Jiawei Pan, Ting Song, and Yao Lu. "Anatomy-Guided Multitask Learning for MRI-Based Classification of Placenta Accreta Spectrum and its Subtypes". In: *2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI)*. 2025, pp. 1–5.

[2] Jiawei Pan, Zilong He, Yue Li, Weixiong Zeng, Yaya Guo, Lixuan Jia, **Hai Jiang**, Weiguo Chen, and Yao Lu. "Atypical architectural distortion detection in digital breast tomosynthesis: a multi-view computer-aided detection model with ipsilateral learning". In: *Physics in Medicine & Biology* 68.23 (2023), p. 235006.

[3] XueFang Wang, **Hai Jiang**, Ruxu Du, Yong Zhong, and Yao Lu. "Anatomical-Prior-Based Multiscale Segmentation of Cardiac Substructures Using Enhanced Skip-Connections and a Triple-View Fusion Network." In: *Submitting* ().

[4] **Hai Jiang**, Chushan Zheng, Jiawei Pan, Yuanpin Zhou, Qiongting Liu, Xiang Zhang, Jun Shen, and Yao Lu. *DECT-based Space-Squeeze Method for Multi-Class Classification of Metastatic Lymph Nodes in Breast Cancer*. 2025. arXiv: `2505.17528 [eess.IV]`. URL: `https://arxiv.org/abs/2505.17528`.

[5] Jiawei Pan, Zilong He, Yue Li, Weixiong Zeng, Yaya Guo, Lixuan Jia, **Hai Jiang**, Weiguo Chen, and Yao Lu. "Multi-view Architectural Distortion Detection with Confidence Boosting in Digital Breast Tomosynthesis". In: *Submitted to MICCAI 2025 (rejected after rebuttal, currently under revision)*.

[6] Qiongting Liu, **Hai Jiang**, Ting Song, and Yao Lu. "Clinical prior-knowledge-based deep learning approach for the diagnosis of placenta accreta spectrum". In: *On-going* ().

## TECHNICAL SKILLS

Programming Python, C/C++, PyTorch, TensorFlow, Keras, MATLAB, LaTeX, Git.

Tools & Platforms Linux (Ubuntu), ONNX, TensorRT, CMake, Microsoft Office, Adobe Photoshop.

Research Methods Deep learning, computer vision, image processing, medical imaging, computational geometry, compressive sensing.

## AWARDS & SCHOLARSHIP

2014–2018 Four-time recipient of the Third-Class Merit Scholarship for Academic Excellence at LZU.

2019–2022 Three-time recipient of the Third-Class Merit Scholarship for Academic Excellence at NKU.

## LANGUAGE PROFICIENCY

Mandarin Native.

English Professional (IELTS 7.0, CET6 476/710, CET4 544/710).

Cantonese Intermediate.

## FUNDING & PATENT WORK

Funding Proposal Writing Contributed to National Key R&D Program of China [No. 2023YFE0204300].

Report Writing Completed three Completion & Progress Reports for National Natural Science Foundation of China [No. 81971691, 12126610] and R&D Program of Pazhou Lab [No. 2023K0606].

Patent Work Completed one patent application.

Medical Device Specification Prepared one medical device application specification.

## TEACHING EXPERIENCE

Courses Taught Calculus, Mathematical Analysis.

Thesis Supervision Breast Cancer Classification Method Based on Dual-Energy CT Images.

## RESEARCH INTERESTS

Artificial Intelligence, Deep Learning (CNNs, GNNs).
Mathematics, Medical Image Analysis, Physics.

## REFERENCES

**Yunhua Xue** Associate Professor, Computational Mathematics, Nankai University
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